

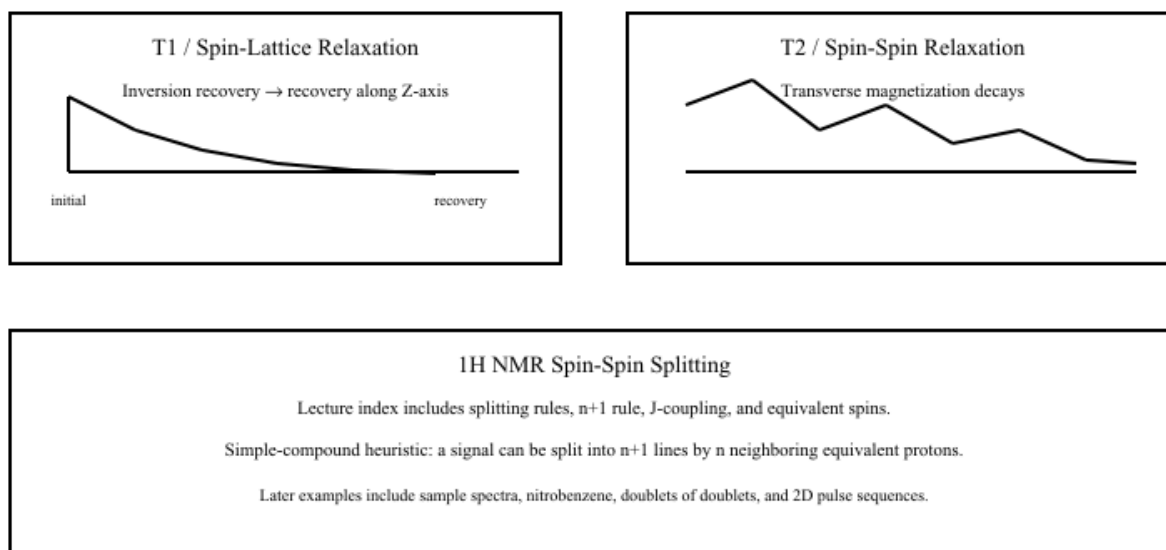
Chemistry Lecture Support Sheet

UCI Chem 131B — Lecture 19: Molecular Structure & Statistical Mechanics

This is an original test/support document created for multimodal RAG evaluation. It is not a transcript or reproduction of the lecture slides.

Source: Rachel Martin, Ph.D., UCI OpenCourseWare. Recorded February 25, 2013. Lecture topic: Spin Rotations T1 & T2. The UCI page provides a topic index and identifies the lecture as Creative Commons licensed.

NMR Relaxation & Spin-Spin Splitting — Study Diagram



Timestamp	Lecture-index topic
03:49	Inversion Recovery (T1)
09:20	Spin-Spin Relaxation (T2)
12:29	Spin Echo (T2)
14:04	1H NMR Spectroscopy: Spin-Spin Splitting
21:18	The n+1 Rule (for simple compounds)
23:49	J-Coupling: Product Basis
32:13	J-Coupling: Equivalent Spins
37:21	Sample Spectrum
38:28	More Complex Splitting Patterns: Nitrobenzene
39:14	Quartet vs. Doublet of Doublets
40:42	NMR Spectrum Example
42:24	General 2D Pulse Sequence

44:53	¹³ C- ¹⁵ N Correlation of Arginine
45:56	H-N HSQC

Use this PDF as a second modality alongside the lecture video. Your system should retrieve relevant PDF concepts together with the matching lecture timestamp/frame.